

ABEE AGDM, AB, Canada

WMO: 712850

Lat: 54.2773N

Lon: 112.9654W

Elev: 664

StdP: 93.60

Time Zone: -7.00 (NAM)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-34.0	-30.1	-37.6	0.1	-33.9	-33.5	0.2	-29.8	9.4	-2.9	8.3	-3.1	1.4	140	0.682

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	27.3	17.3	25.6	16.5	23.9	15.7	19.0	25.1	17.8	23.6	16.7	22.1	3.7	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.6	12.8	22.2	15.5	11.9	20.7	14.4	11.1	19.3	56.8	25.1	52.9	23.8	49.4	22.1	23.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.6	7.5	6.6	DB	-39.6	30.1	3.6	1.4	-42.2	31.1	-44.3	32.0	-46.3	32.7	-48.9	33.8	(4)
(5)			WB	-39.7	21.5	3.6	1.6	-42.3	22.6	-44.4	23.5	-46.4	24.4	-49.0	25.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	1.4	-12.9	-12.6	-6.1	2.6	9.4	13.5	16.0	14.3	9.4	2.6	-6.6	-13.2
	DBStd	12.26	8.51	7.94	7.44	5.30	4.44	2.95	2.70	3.20	4.30	4.73	6.99	7.88
	HDD10.0	3653	711	633	499	225	64	6	0	4	62	232	499	718
	HDD18.3	6183	969	866	758	471	277	147	83	128	270	487	749	977
	CDD10.0	530	0	0	0	4	47	111	186	138	43	3	0	0
	CDD18.3	17	0	0	0	0	1	2	10	4	1	0	0	0
	CDH23.3	473	0	0	0	1	37	66	201	123	44	2	0	0
	CDH26.7	78	0	0	0	0	4	7	40	19	8	0	0	0
Wind	WSAvg	3.0	2.8	2.8	3.2	3.7	3.5	3.1	2.8	2.6	2.8	3.1	2.9	2.6
Precipitation	PrecAvg	405	17	12	16	24	41	70	78	52	33	18	16	17
	PrecMax	507	39	27	34	57	86	138	155	90	74	56	41	39
	PrecMin	300	0	0	3	3	6	22	22	21	9	5	5	0
	PrecStd	61	11	7	9	13	19	30	27	19	18	11	9	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.6	6.6	9.9	21.8	26.9	27.5	30.1	29.0	27.7	21.8	11.2	5.9
		MCWB	2.7	2.5	4.7	10.5	13.8	16.3	20.1	17.6	16.0	11.8	6.4	2.5
	2%	DB	3.9	3.5	7.2	17.9	23.9	25.2	27.6	26.5	24.1	16.0	6.5	3.0
		MCWB	1.1	0.6	3.1	8.7	12.4	15.6	18.8	17.4	14.9	9.1	3.1	0.5
	5%	DB	1.8	1.3	5.1	14.6	21.4	23.3	25.5	24.4	21.1	13.0	4.3	0.6
		MCWB	-0.2	-0.7	1.8	6.8	11.2	14.6	17.7	16.5	13.4	7.1	1.8	-1.1
	10%	DB	-0.5	-0.8	3.4	11.7	19.4	21.2	23.7	22.6	18.1	10.6	2.0	-1.8
		MCWB	-2.0	-2.3	0.8	5.5	10.2	13.6	16.8	15.9	11.8	5.8	0.2	-3.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.0	2.7	5.2	11.0	14.7	18.2	21.5	20.0	17.2	12.4	6.3	2.6
		MCDB	6.1	6.1	8.7	20.7	23.9	25.1	27.7	25.5	25.1	20.9	10.6	5.3
	2%	WB	1.4	0.8	3.3	8.8	13.2	16.7	19.7	18.4	15.4	9.8	3.6	0.6
		MCDB	3.6	3.0	6.6	17.0	21.5	22.6	26.0	24.5	22.7	15.3	6.0	2.7
	5%	WB	-0.1	-0.7	2.0	7.2	12.0	15.5	18.5	17.3	13.9	7.7	1.9	-1.1
		MCDB	1.9	1.2	4.8	13.6	19.7	21.5	24.3	23.0	20.6	12.3	3.9	0.6
	10%	WB	-2.0	-2.4	0.8	5.7	10.9	14.5	17.3	16.2	12.4	6.3	0.2	-3.0
		MCDB	-0.6	-0.8	3.1	11.2	17.9	19.9	22.5	21.8	17.3	9.8	1.8	-1.8
Mean Daily Temperature Range		MDBR	10.8	11.9	10.8	11.9	13.9	12.5	12.9	13.5	13.5	11.6	9.4	10.5
	5% DB	MCDBR	12.2	14.3	12.1	17.6	18.4	16.8	16.7	17.4	19.7	16.5	11.9	12.4
		MCWBR	9.8	11.8	8.7	9.5	8.6	8.4	8.9	9.3	10.8	10.2	8.6	10.4
	5% WB	MCDBR	11.8	13.8	11.5	16.3	16.4	14.4	15.2	15.5	17.9	15.8	11.0	12.3
		MCWBR	9.7	11.5	8.3	9.4	8.0	7.7	8.6	8.8	10.0	10.0	8.3	10.4
Clear-Sky Solar Irradiance	taub	0.253	0.271	0.305	0.325	0.338	0.352	0.351	0.359	0.305	0.280	0.249	0.235	
	taud	2.280	2.295	2.304	2.386	2.406	2.378	2.430	2.354	2.516	2.557	2.455	2.289	
	Ebn at Noon	719	831	876	905	908	895	889	857	875	829	743	669	
	Edh at Noon	56	78	98	104	108	113	105	107	79	61	47	44	
All-Sky Solar Radiation	RadAvg	0.86	1.89	3.40	4.66	5.68	5.95	6.02	4.93	3.40	1.92	0.90	0.63	
	RadStd	0.10	0.14	0.26	0.39	0.46	0.40	0.32	0.35	0.35	0.25	0.10	0.05	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page